

IoT WEEK 8

COLOR MEMORY NOTES

Tomorrow's exam - fast recall + 8-mark focus

IEEE 802.15.4 + ZigBee

COLOR CODE = MEMORY CODE

BLUE

Core definitions / basic facts

GREEN

Features / benefits / easy-scoring points

YELLOW

Numbers, keywords, must-memorize facts

ORANGE

Comparisons / differences / warning points

PURPLE

Memory tricks / mnemonics

Emergency Priority

- 1) IEEE 802.15.4 features 2) Beacon vs Non-Beacon 3) ZigBee features
4) ZigBee Mesh 5) Device types 6) AODV routing

Use colors to recall categories, not full paragraphs.

1. IEEE 802.15.4 - FAST MEMORY MAP

High priority: definitions, modulation, access method, range and topology

CORE FEATURES

- Low data-rate WPAN
- Low-power / long battery life
- Mainly PHY + MAC layers
- Works in ISM band
- Noise/interference tolerant
- Star and Mesh topology

MUST-MEMORIZE TECH

- DSSS modulation
- Low speed -> BPSK
- High speed -> O-QPSK
- Channel access -> CSMA-CA
- Duty cycle < 1%
- Min power = -3 dBm = 0.5 mW

NUMBERS = EASY MARKS

Normal range: 10-75 m

Outdoor best: up to 1000 m

Duty cycle: <1%

Tip: write exact numbers in an 8-mark answer - they make the answer look strong.

MEMORY TRICK

Low Power Student Drinks Beer Quickly, Comes Home

L=Low data | P=PHY/MAC | S=ISM | D=DSSS | B=BPSK | Q=O-QPSK | C=CSMA-CA

2. IEEE 802.15.4 - COMPARISONS

Beacon vs Non-Beacon + FFD/RFD + variants + frame types

BEACON vs NON-BEACON

BEACON ENABLED

- Periodic beacon transmission
- Slotted CSMA/CA
- Superframe structure
- PAN coordinator manages
- Synchronization + association

NON-BEACON

- Unslotted CSMA/CA
- Contention based
- Beacon mainly for discovery
- Source + destination IDs
- More decentralized

BEACON MEMORY

BSS = Beacon - Slotted - Sync

NON-BEACON MEMORY

USD = Unslotted - IDs - Decentralized

FFD vs RFD

FFD = Full Function Device

- Talks to all device types
- Supports full protocol

RFD = Reduced Function Device

- Talks only to FFD
- Low power + low CPU/RAM

MICRO FACTS

Variants: A/B Base | C China | D Japan | E Industrial | F Active RFID | G Smart Grid

Frame memory: BMCAD = Beacon - MAC - Command - Acknowledgement - Data

Memory: "Bro My Cat Ate Dinner" = B M C A D

3. ZigBee - FEATURES + ARCHITECTURE

Think: IEEE 802.15.4 gives lower layers; ZigBee adds smart upper-layer functions

CORE IDEA

ZigBee is a widely used enhancement of IEEE 802.15.4 for low-power IoT and sensor networks.

LAYER MEMORY

Application / Upper

Network (ZigBee)

MAC (802.15.4)

PHY (802.15.4)

ZigBee = AERM

- Authentication of valid nodes
- Encryption for security
- Routing and forwarding
- Mesh networking support
- Popular in wireless sensor networks

AERM

Authentication - Encryption - Routing - Mesh

IMPORTANT COMPONENTS

ZDO

Device management + security + policies

Think: ZDO = device boss

APS

Interfacing/control services

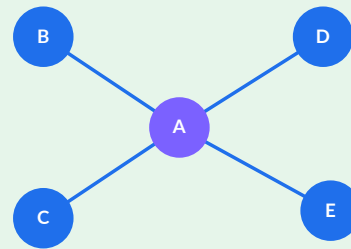
Bridge between network and upper layers

4. ZigBee MESH + DEVICE TYPES

Very high-yield: topology, self-healing, coordinator/router/end-device roles

MESH = RELAY + RELIABILITY

- Node in range -> direct communication
- Node out of range -> message relayed by intermediate nodes
- Supports large-area deployment
- Alternative paths increase reliability
- Self-configuring + self-healing



MESH MEMORY

CRLSH = Communicate - Relay - Large area - Self-configuring - Healing

DEVICE TYPES = C - R - E

ZC Coordinator

- One per network
- Starts network
- Root / bridge
- Trust center + keys

ZR Router

- Runs applications
- Relays information
- Extends connectivity

ZED End Device

- Talks to parent
- Cannot relay
- Sleeps often
- Low cost + long battery

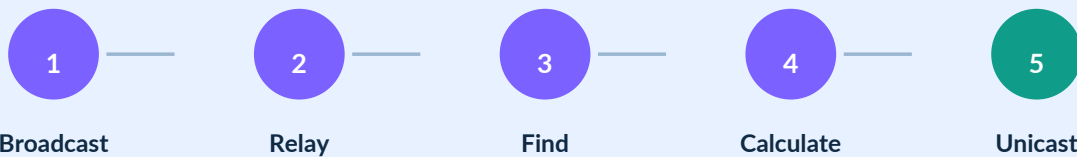
DEVICE MEMORY

C-R-E = Create network - Relay data - End/sleep

5. AODV ROUTING + APPLICATIONS

Use the flow below to write a clean 8-mark explanation

AODV = Ad Hoc On-Demand Distance Vector



AODV MEMORY

BRFCU = Broadcast - Relay - Find - Calculate - Unicast

ZigBee APPLICATIONS

● Building automation

● Smart energy monitoring

● Home automation

● Telecom services

● Remote control / consumer electronics

● Health & fitness monitoring

● LED lighting control

Exam trick:

For 8 marks: Definition -> 5-7 points -> memory keyword/diagram -> one-line conclusion.

6. FINAL 5-MINUTE REVISION

Read this page three times just before entering the exam hall

IEEE 802.15.4

- Low-rate WPAN
- PHY + MAC
- ISM band
- DSSS
- BPSK low speed
- O-QPSK high speed
- CSMA-CA
- 10-75 m normal
- up to 1000 m outdoor
- Star + Mesh

ZigBee

- Enhancement of IEEE 802.15.4
- Layer 3 and above
- Authentication + Encryption
- Routing + Forwarding
- Mesh support
- Wireless sensor networks

MEMORY BANK

BSS	Beacon - Slotted - Sync	CRE	Coordinator - Router - End Device
USD	Unslotted - IDs - Decentralized	BRFCU	Broadcast - Relay - Find - Calculate - Unicast
AERM	Authentication - Encryption - Routing - Mesh		

LAST RULE: Do not memorize paragraphs. Memorize keywords + order + 1 example/number.